



Q-MAT: Quantum Multi-Asset Terminal

Technical Overview and Challenge Deliverables Mapping

Team: Vishal Dhanure, Lumbini Devi, and Yamini

Repository: github.com/Vi1234sh12/vanguard-quantum-portfolio

Abstract

Q-MAT (Quantum Multi-Asset Terminal) is a full-stack, hybrid quantum–classical architecture for institutional multi-asset portfolio construction. The system translates a guarded Markowitz mean–variance problem into a 78-variable QUBO/Ising formulation that is compatible with QAOA and annealing-based solvers. In the current prototype, a simulated-annealing candidate is passed to a constrained SLSQP “polisher.” The resulting portfolio is accepted only after explicit checks of budget, position, sector, turnover, stress, and liquidity limits. This architecture combines discrete global search, exact classical feasibility enforcement, out-of-sample validation, and an interactive compliance-oriented interface.

1 Mathematical Formulation

1.1 Notation and basic portfolio quantities

The core symbols used throughout the formulation are summarized below.

Symbol	Meaning
$P_{i,t}$	closing price of asset i at time t
$r_{i,t}$	one-period return of asset i
w_i	portfolio weight assigned to asset i
μ_i	annualized expected return of asset i
Σ	annualized asset-return covariance matrix
$w^{(0)}$	current or pre-trade portfolio weights
λ, γ	risk-aversion and trading-regularization coefficients
ϵ	requested minimum expected return

The simple return of asset i is

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} = \frac{P_{i,t}}{P_{i,t-1}} - 1.$$

For T daily observations and annualization factor $A = 252$, the sample estimates are

$$\hat{\mu} = \frac{A}{T} \sum_{t=1}^T r_t, \quad \hat{\Sigma} = \frac{A}{T-1} \sum_{t=1}^T (r_t - \bar{r})(r_t - \bar{r})^\top.$$

Q-MAT replaces the noisy sample-mean vector with Black–Litterman expected returns, while retaining a regularized factor estimate of covariance. For portfolio weights w , the basic risk–return quantities are

$$\mu_p = \mu^\top w, \quad \sigma_p^2 = w^\top \Sigma w, \quad \sigma_p = \sqrt{w^\top \Sigma w}, \quad \text{Sharpe}(w) = \frac{\mu_p - r_f}{\sigma_p}.$$

For N investable assets, let $w \in \mathbb{R}^N$ denote portfolio weights, μ the Black–Litterman expected-return vector, Σ the PCA factor-model covariance matrix, and $w^{(0)}$ the initial portfolio. Following the Markowitz mean–variance framework [1], Q-MAT minimizes

$$\mathcal{O}(w) = \lambda w^\top \Sigma w - \mu^\top w + \gamma \sum_{i=1}^N \left(w_i - w_i^{(0)} \right)^2.$$

where $\lambda > 0$ controls risk aversion and $\gamma \geq 0$ penalizes trading away from the current allocation.

The exact portfolio turnover and a simple proportional trading-cost model are

$$\text{TO}(w) = \sum_{i=1}^N |w_i - w_i^{(0)}|, \quad C_{\text{trade}}(w) = \sum_{i=1}^N c_i |w_i - w_i^{(0)}|,$$

where c_i is the round-trip cost per unit weight. The quadratic term in $\mathcal{O}(w)$ is a smooth trading regularizer; the exact L_1 expression is used for the final turnover check.

Each weight is represented with six binary variables:

$$w_i = \Delta_i \sum_{j=0}^5 2^j x_{i,j}, \quad x_{i,j} \in \{0, 1\}.$$

This gives 64 levels, indexed from 0 through 63. For a 25% position cap, setting $\Delta_i = 0.25/63$ gives a grid spacing of approximately 0.397 percentage points. Thirteen instruments—12 anonymized ETFs and one protective-put instrument—therefore require $13 \times 6 = 78$ binary variables.

The institutional guardrails are

$$\begin{aligned} \sum_{i=1}^N w_i &= 1, & (\text{budget}) \\ 0 \leq w_i &\leq 0.25, & (\text{position limits}) \\ \sum_{i \in \mathcal{S}_k} w_i &\leq 0.40 \quad \forall k, & (\text{sector limits}) \\ \sum_{i=1}^N |w_i - w_i^{(0)}| &\leq 0.15, & (\text{turnover}) \\ \sum_{i \in \mathcal{E}} w_i &\leq 0.40, & (\text{equity stress cap}) \\ w_i &\leq \ell_i \quad \forall i, & (\text{liquidity limits}), \end{aligned}$$

where $\mathcal{S}_k$ is sector k , $\mathcal{E}$ is the equity subset, and ℓ_i is the maximum portfolio weight permitted by tradable-volume rules.

The requested return goal can be written as

$$\mu^\top w \geq \epsilon.$$

It is enforced when feasible under the hard institutional guardrails. If no feasible portfolio reaches ϵ , Q-MAT keeps the guardrails fixed, reports the target as infeasible, and returns the highest-return feasible candidate rather than silently relaxing risk or compliance limits.

2 Quantum-Compatible QUBO and Ising Formulation

After substituting the binary weight expansion, the objective and quadratic constraint penalties are assembled into

$$H(x) = \mathcal{O}(w(x)) + P_{\text{bud}} \left(\sum_i w_i(x) - 1 \right)^2 + P_{\text{ret}} [\mu^\top w(x) - \epsilon]^2 + \dots,$$

with the sign or slack-variable construction of the return term selected to penalize only target shortfalls. The resulting QUBO has the standard form

$$\min_{x \in \{0,1\}^{78}} x^\top Q x.$$

Inequalities that cannot be represented exactly by a simple squared equality are encoded with slack variables or treated as soft candidate-generation penalties; all hard limits are subsequently enforced by the classical polishing stage.

The 78-variable count refers to the six allocation bits for each of the 13 instruments. If an inequality is converted to an equality with binary slack variables, those ancillary bits increase the total logical-variable count. The current hybrid design therefore treats 78 as the core allocation dimension and relies on the classical stage for exact enforcement of constraints that would otherwise require many additional ancillas.

Writing the binary encoding compactly as $w = Ax$, the quadratic risk and turnover terms become

$$\lambda x^\top A^\top \Sigma A x + \gamma (Ax - w^{(0)})^\top (Ax - w^{(0)}).$$

Linear terms are placed on the diagonal because $x_j^2 = x_j$ for binary variables. For example, the budget penalty

$$P_{\text{bud}} (\mathbf{1}^\top Ax - 1)^2$$

contributes both linear biases and pairwise couplings to Q .

2.1 Dynamic spectral penalty scaling

Penalty magnitudes are initialized relative to the covariance scale. In particular, Q-MAT computes the largest eigenvalue $\lambda_{\max}(\Sigma)$ and selects penalty coefficients above the corresponding objective-energy scale. The coefficients are then calibrated to account for the binary encoding and linear-return term. This avoids penalties that are too weak to influence feasibility or so large that they erase useful objective resolution.

2.2 Ising mapping

Following the standard QUBO–Ising construction [2], binary variables are mapped to Pauli-Z spin variables using

$$x_i = \frac{1 - Z_i}{2}, \quad Z_i \in \{-1, +1\}.$$

For a symmetric QUBO convention $x^\top Q x = \sum_i Q_{ii} x_i + 2 \sum_{i < j} Q_{ij} x_i x_j$, the cost Hamiltonian is

$$H_C = c + \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j,$$

where

$$J_{ij} = \frac{Q_{ij}}{2}, \quad h_i = -\frac{Q_{ii}}{2} - \frac{1}{2} \sum_{j \neq i} Q_{ij},$$

and c is an optimization-independent constant. These coefficients can drive a gate-model QAOA circuit or an Ising/annealing solver. The demonstrator uses D-Wave Ocean’s **neal** simulated-annealing sampler; **neal** is a classical QUBO baseline rather than a physical quantum processor.

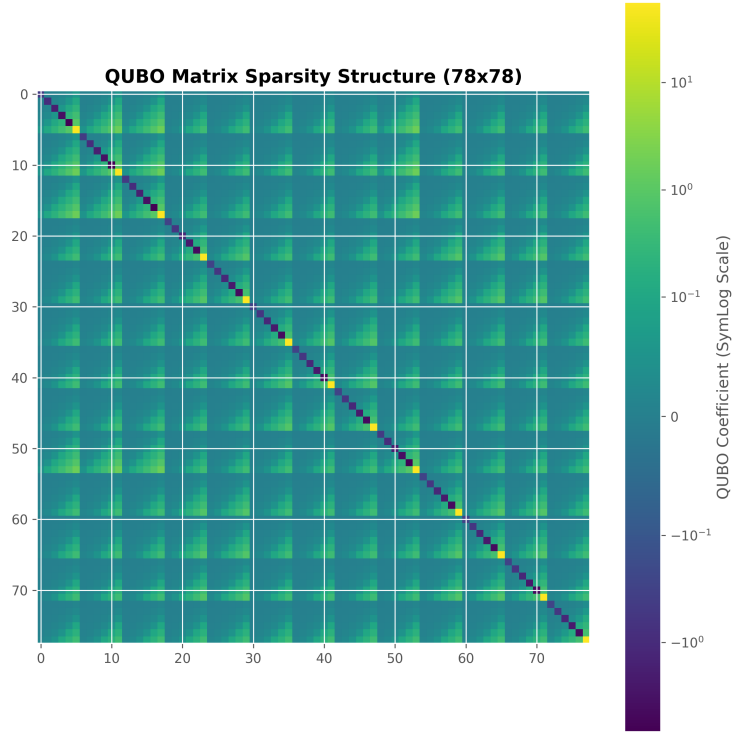


Figure 1: Sparsity and coefficient structure of the 78×78 QUBO matrix. The logarithmic color scale exposes both dominant diagonal terms and weaker cross-variable couplings.

2.3 QAOA workflow

In a gate-model implementation, QAOA [3] prepares and iteratively improves a parameterized quantum state as follows:

1. **Initialize the quantum state.** Prepare the n -qubit uniform superposition $|+\rangle^{\otimes n}$, where n is the number of binary decision variables.

2. **Apply alternating unitaries.** At each QAOA layer, apply the problem unitary

$$U_P(\gamma_k) = e^{-i\gamma_k H_C}$$

to encode the portfolio cost, followed by the mixer unitary

$$U_M(\beta_k) = e^{-i\beta_k \sum_{j=1}^n X_j}$$

to explore alternative bitstrings. For depth p , this produces $|\psi(\gamma, \beta)\rangle = \prod_{k=1}^p U_M(\beta_k)U_P(\gamma_k)|+\rangle^{\otimes n}$.

3. **Measure the cost.** Estimate the objective expectation $\langle H_C \rangle = \langle \psi | H_C | \psi \rangle$ from repeated circuit measurements.
4. **Update the parameters.** A classical Adam optimizer uses the measured cost to update γ and β .
5. **Repeat until convergence.** Re-execute the parameterized circuit until the stopping criterion is met, then decode the best measured bitstring as the candidate portfolio.

QUANTUM PROCESSING

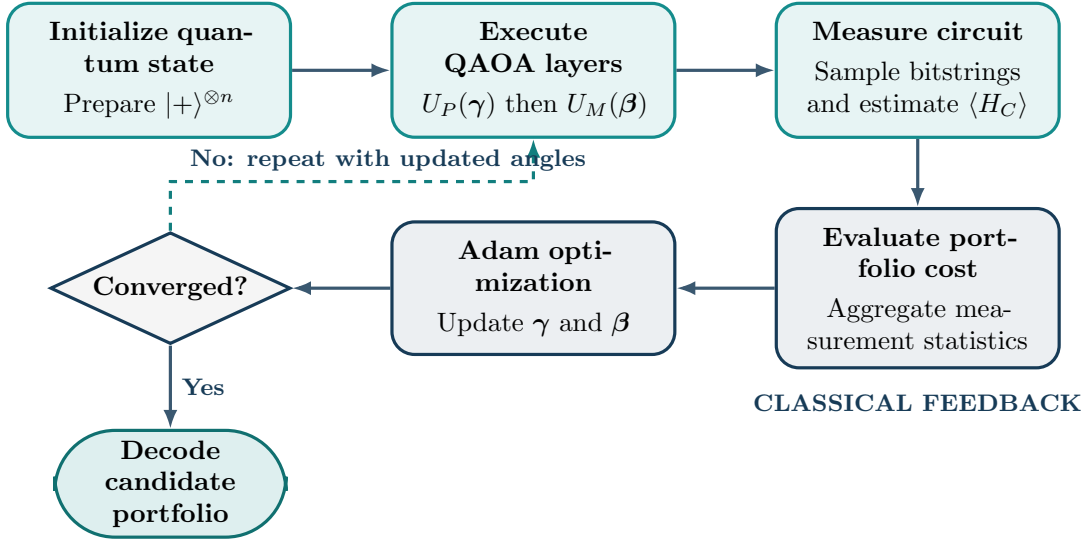
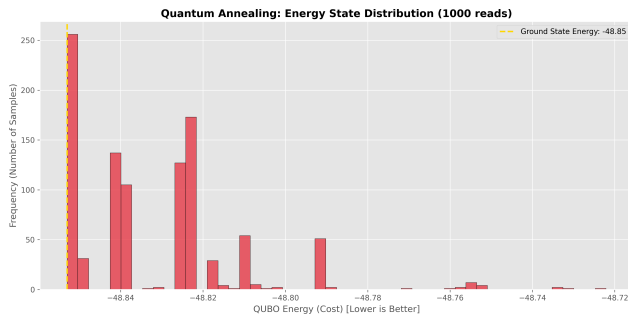
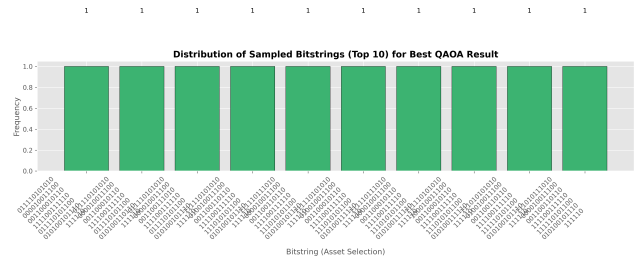


Figure 2: QAOA quantum–classical feedback loop. Quantum circuit execution produces cost estimates; Adam updates the variational angles until the convergence test releases a candidate portfolio.



(a) Sampled QUBO-energy distribution and ground-state reference.



(b) Most frequently observed candidate bitstrings.

Figure 3: QUBO-search diagnostic views used to inspect energy concentration and solution diversity.

3 Data and Financial Modeling Strategy

Q-MAT uses five years of historical daily price data for 12 multi-asset ETFs, retrieved from Yahoo Finance through the open-source `yfinance` Python library [6]. User-facing outputs remove ticker symbols and relabel the instruments `Asset_1` through `Asset_12`. The first four years form the estimation set, while the final year is held out for out-of-sample testing.

The modeling pipeline comprises:

- **PCA factor covariance.** Latent factors produce a regularized covariance estimate $\Sigma = B\Sigma_F B^\top + D$, reducing sampling noise relative to an unrestricted sample covariance.
- **Monte Carlo views.** One thousand geometric-Brownian-motion paths generate forward-looking return scenarios.
- **Black–Litterman returns.** Scenario-derived views are blended with reverse-optimized equilibrium returns $\Pi = \delta\Sigma w_{\text{mkt}}$, reducing the sensitivity of unconstrained Markowitz allocations to noisy mean estimates.
- **Protective put.** A one-year European protective put is valued with Black–Scholes and represented as a thirteenth candidate instrument so that the optimizer can compare explicit hedging cost with downside protection.

3.1 Core financial-model equations

For the PCA factor model, returns are approximated by

$$r_t = Bf_t + \varepsilon_t, \quad \Sigma = B\Sigma_F B^\top + D,$$

where B contains the retained principal-component loadings, Σ_F is factor covariance, and D is diagonal residual covariance.

The Monte Carlo scenarios follow geometric Brownian motion,

$$dS_t = \mu_S S_t dt + \sigma_S S_t dW_t,$$

with discrete simulation step

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu_S - \frac{1}{2} \sigma_S^2 \right) \Delta t + \sigma_S \sqrt{\Delta t} Z_t \right], \quad Z_t \sim \mathcal{N}(0, 1).$$

Black–Litterman [4] combines equilibrium returns $\Pi = \delta\Sigma w_{\text{mkt}}$ with views q through

$$\mu_{\text{BL}} = \left[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P \right]^{-1} \left[(\tau\Sigma)^{-1} \Pi + P^\top \Omega^{-1} q \right],$$

where P selects the viewed assets, Ω represents view uncertainty, and τ scales prior uncertainty.

The Black–Scholes model [5] gives the value of a European put as

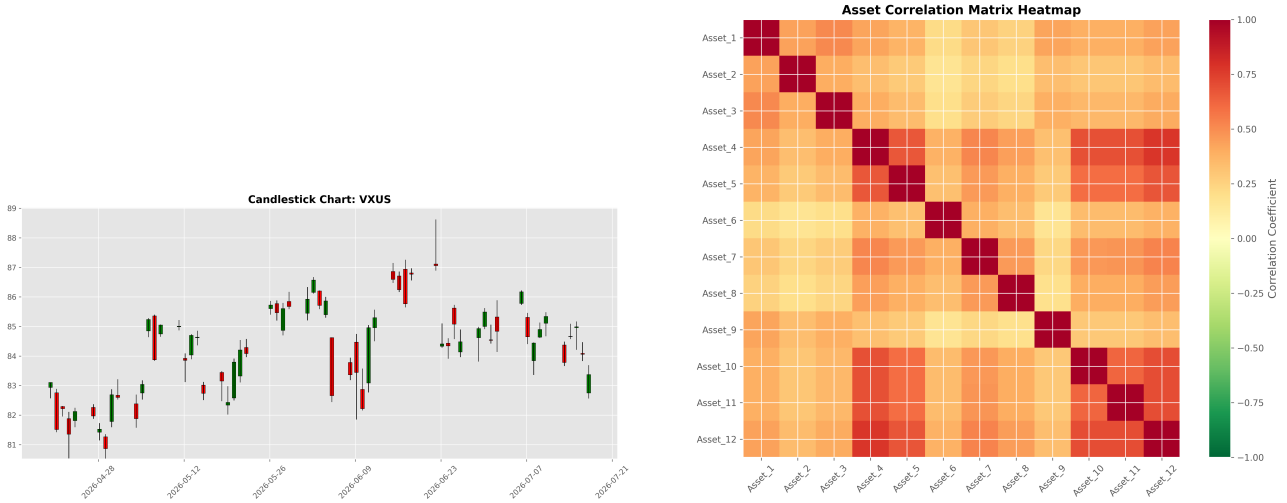
$$P_0 = K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1),$$

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

At expiry, the protected position has gross payoff

$$S_T + \max(K - S_T, 0) = \max(S_T, K),$$

before subtracting the option premium and trading costs.



(a) Illustrative market-price series used for return estimation. (b) Cross-asset correlation structure for the anonymized universe.

Figure 4: Market-data and dependence diagnostics supporting the financial-modeling pipeline.

4 Baseline, Hybrid Solver, and Feasibility

The classical benchmark is a strict long-only minimum-variance quadratic program implemented in CVXPY. Q-MAT’s hybrid solver then operates in two stages:

The benchmark solves

$$\min_w w^\top \Sigma w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0,$$

with the applicable position, sector, turnover, stress, and liquidity limits added for like-for-like comparison.

1. **Discrete candidate generation.** The QUBO/Ising solver searches the encoded allocation space. A warm-start bias favors states near $w^{(0)}$, improving the likelihood that sampled candidates lie near the turnover-feasible region.
2. **Constrained polishing.** The best decoded candidate initializes SLSQP, which optimizes the continuous objective subject to the original hard constraints. A differentiable Pseudo-Huber approximation,

$$\phi_\delta(z) = \delta^2 \left(\sqrt{1 + (z/\delta)^2} - 1 \right),$$

provides a smooth turnover surrogate during optimization. Final acceptance is based on the exact L_1 turnover constraint stated above, not on the surrogate alone.

Thus, the QUBO stage proposes a strong starting point, while the classical stage and post-solve validator enforce feasibility. “Zero breaches” means that every reported final portfolio passes numerical-tolerance checks against all original hard constraints.

5 Tunable Investment Goals

The dashboard exposes a *Tunable Goals* panel with:

- profile presets: Conservative, Balanced, and Aggressive;
- risk aversion λ from 1 to 10;
- minimum return target ϵ from 2% to 12%; and
- maximum turnover from 5% to 50%.

Selecting *Execute Pipeline* reruns the asynchronous workflow with the new parameters and displays how the allocation and active constraints change.

6 Outputs, Comparison, and Explainability

The interface compares three portfolio states: the classical minimum-variance benchmark, the raw QUBO candidate, and the final hybrid portfolio. A dual-axis chart reports expected return and volatility, while compliance gauges show budget, turnover, equity stress-cap, and liquidity utilization. The intended result is to retain useful return discovered during discrete search while the SLSQP stage removes violations and reduces avoidable risk.

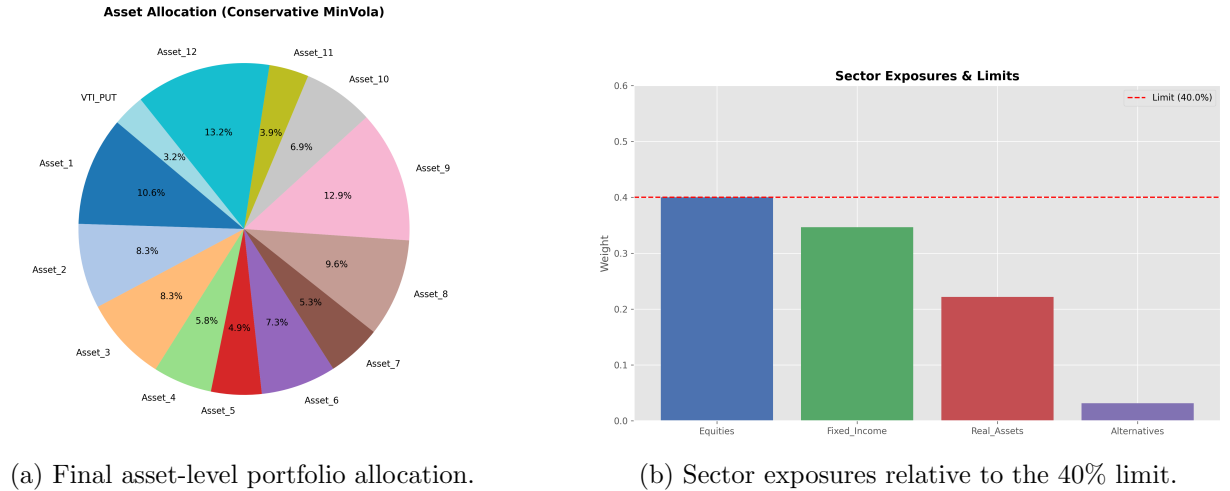


Figure 5: Allocation and compliance views for interpreting the optimized portfolio.

An AI Co-Pilot converts optimization outputs into a concise executive memo. It reports the rationale for material positions, identifies binding guardrails, and summarizes expected annualized return. If a requested target is infeasible, the memo explicitly distinguishes the target from the maximum feasible return; for example:

The requested minimum return of 7.00% was infeasible under the selected liquidity and turnover limits. The optimizer retained zero hard-constraint breaches and returned the highest feasible solution found, with an expected return of 3.50%.

7 Classical and Optimization Validation

The evaluation is organized around three research questions:

- **RQ1—Search efficacy:** Does QUBO-guided annealing find lower-energy allocations than uninformed random sampling?
- **RQ2—Portfolio utility:** Does the hybrid candidate improve risk-adjusted performance relative to equal-weight and classical baselines?
- **RQ3—Institutional viability:** Can the final allocation satisfy every hard guardrail while remaining computationally tractable?

Q-MAT benchmarks candidate portfolios with SLSQP and NSGA-II. The application plots the hybrid result against the classical efficient frontier and Capital Allocation Line, allowing the user to inspect its risk–return position rather than relying on a single objective value. Figure 6 maps each research question to its experiment and evidence.

The efficient frontier is generated by solving a sequence of target-return problems,

$$\min_w w^\top \Sigma w \quad \text{subject to} \quad \mu^\top w \geq \rho,$$

for increasing targets ρ . NSGA-II instead treats risk and return as the two-objective problem

$$\min_w \left(w^\top \Sigma w, -\mu^\top w \right),$$

subject to the same feasible region.

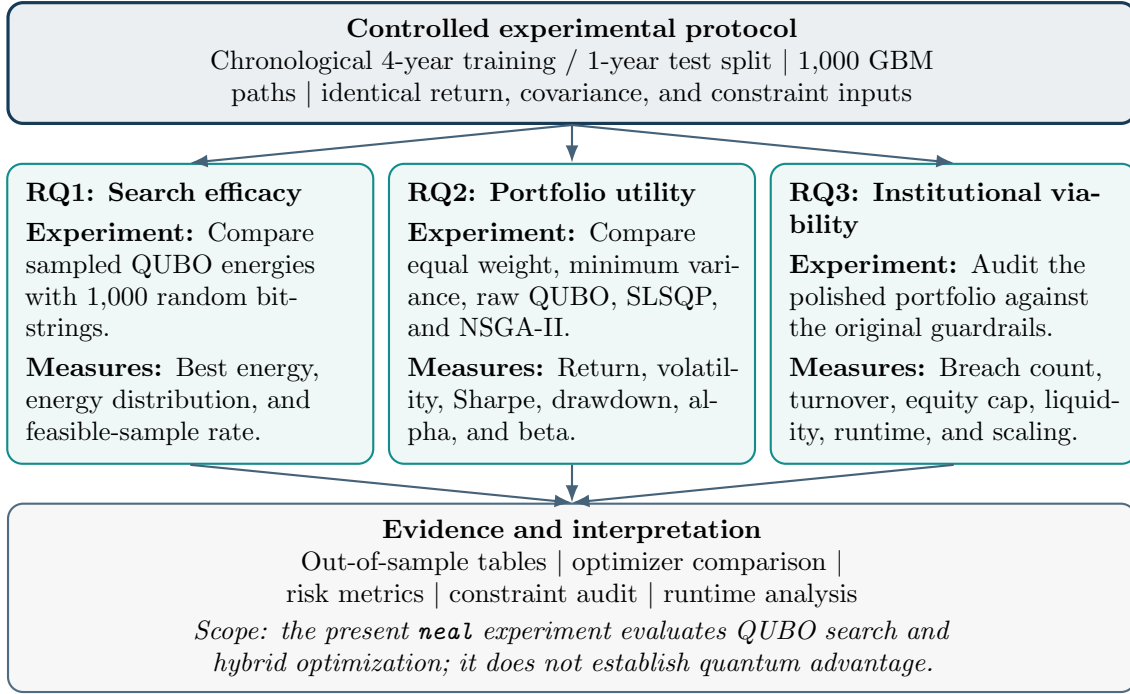
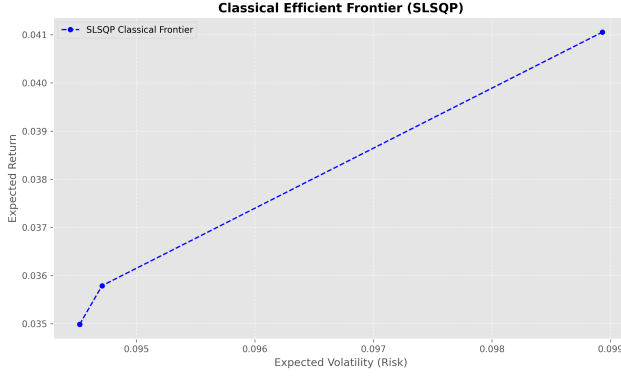
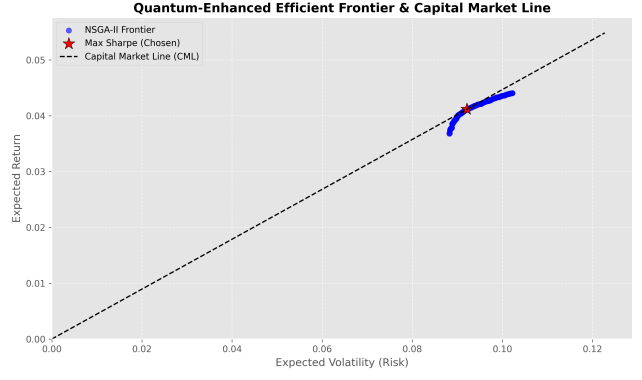


Figure 6: Scientific validation framework linking the research questions to controlled comparisons, quantitative measures, and defensible conclusions.

Optimization effectiveness is tested by comparing annealer QUBO energies with 1,000 randomly sampled bitstrings. The energy distributions, best-observed energy, feasibility rate, and repeated-run variability are reported together. Concentration at lower energy than the random baseline provides evidence of effective QUBO search; it does not, by itself, establish quantum advantage.



(a) Classical SLSQP efficient frontier.



(b) NSGA-II frontier, maximum-Sharpe point, and Capital Market Line.

Figure 7: Classical and multi-objective risk–return frontiers used to benchmark the hybrid candidate.

8 Interactive Prototype

The prototype is an asynchronous Flask/PostgreSQL application with the following demonstration flow:

8.1 Research architecture

Figure 8 summarizes the end-to-end research architecture. Financial estimation and scenario generation feed a discrete QUBO/Ising model; the sampled allocation is then polished against the original continuous constraints before validation and presentation.

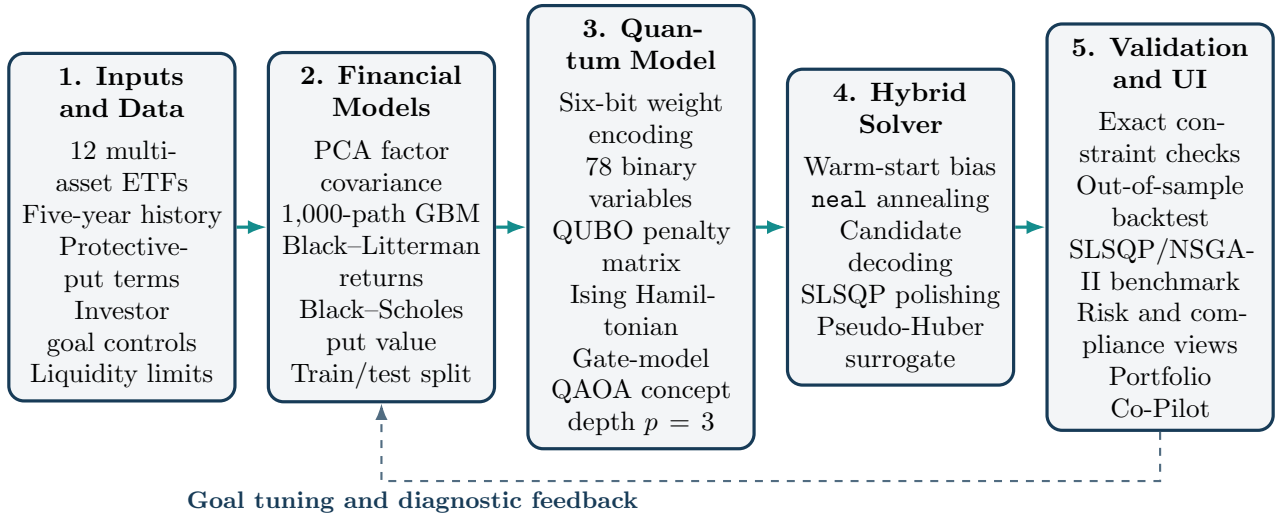


Figure 8: Q-MAT research architecture. The discrete solver generates a portfolio candidate; SLSQP and the exact post-solve validator determine whether the reported allocation satisfies the institutional guardrails.

1. The user adjusts goal parameters and starts the pipeline.
2. A live progress indicator reports Data $\rightarrow$ Black–Litterman $\rightarrow$ QUBO $\rightarrow$ Anneal $\rightarrow$ Polish.

3. The application opens a three-tab inspection terminal:

- **Overview:** headline metrics, method comparison, efficient frontier, allocations, and Co-Pilot memo;
- **Quantum Architecture:** circuit viewer, QUBO heatmap, rendered equations, and annealer-versus-random validation; and
- **Risk & Compliance:** viability score, constraint gauges, drawdown curve, and correlation matrix.

8.2 Prototype screenshots

Q-MAT (Quantum Multi-Asset Terminal)

vanguard-challenge | hybrid-qaoa-architecture

// EXECUTION HISTORY						36 RECORDS
ID	PROFILE NAME	STATUS	EXP. RETURN	VOLATILITY	SHARPE	ACTION
#36	Conservative MinVola	COMPLETED	3.96%	10.25%	0.3859	INSPECT -
#35	Conservative MinVola	COMPLETED	3.57%	9.8%	0.3645	INSPECT -
#34	Aggressive Growth	COMPLETED	3.93%	10.22%	0.3848	INSPECT -
#33	Balanced MinVola	COMPLETED	3.57%	9.8%	0.3645	INSPECT -
#32	Conservative MinVola	COMPLETED	3.57%	9.8%	0.3645	INSPECT -
#31	Aggressive Growth	COMPLETED	3.57%	9.8%	0.3645	INSPECT -
#30	Balanced MinVola	COMPLETED	3.57%	9.8%	0.3645	INSPECT -
#29	Conservative MinVola	COMPLETED	3.38%	9.48%	0.3566	INSPECT -
#28	Vanguard Conservative Run	COMPLETED	3.57%	9.8%	0.3645	INSPECT -
#27	Vanguard Conservative Run	COMPLETED	3.47%	9.79%	0.3546	INSPECT -
#26	Vanguard Conservative Run	COMPLETED	3.47%	9.79%	0.3546	INSPECT -
#25	Vanguard Conservative Run	COMPLETED	3.47%	9.79%	0.3546	INSPECT -
#24	Vanguard Conservative Run	COMPLETED	3.52%	9.79%	0.359	INSPECT -
#23	Vanguard Conservative Run	COMPLETED	3.52%	9.79%	0.359	INSPECT -
#22	Vanguard Conservative Run	COMPLETED	3.52%	9.79%	0.359	INSPECT -
#21	Vanguard Conservative Run	COMPLETED	3.5%	9.8%	0.3577	INSPECT -
#20	Vanguard Conservative Run	COMPLETED	3.5%	9.8%	0.3577	INSPECT -
#19	Vanguard Conservative Run	COMPLETED	3.5%	9.8%	0.3577	INSPECT -
#18	Vanguard Conservative Run	COMPLETED	3.5%	9.8%	0.3577	INSPECT -
#17	Vanguard Conservative Run	COMPLETED	3.5%	9.8%	0.3577	INSPECT -
#16	Vanguard Conservative Run	COMPLETED	3.9%	9.83%	0.3968	INSPECT -
#15	Vanguard Conservative Run	COMPLETED	3.89%	9.84%	0.3953	INSPECT -
#14	Vanguard Conservative Run	COMPLETED	3.5%	9.8%	0.3577	INSPECT -
#13	Vanguard Conservative Run	COMPLETED	3.5%	9.8%	0.3577	INSPECT -
#12	Vanguard Conservative Run	COMPLETED	3.48%	9.74%	0.3541	INSPECT -
#11	Vanguard Conservative Run	COMPLETED	3.53%	9.79%	0.3617	INSPECT -
#10	Vanguard Conservative Run	COMPLETED	2.89%	8.89%	0.3252	INSPECT -
#9	Vanguard Conservative Run	COMPLETED	2.8%	9.62%	0.3102	INSPECT -
#8	Vanguard Conservative Run	COMPLETED	2.8%	9.62%	0.3102	INSPECT -
#7	Vanguard Conservative Run	COMPLETED	4.16%	10.29%	0.4841	INSPECT -
#6	Vanguard Conservative Run	COMPLETED	4.16%	10.29%	0.4841	INSPECT -
#5	Vanguard Conservative Run	COMPLETED	4.16%	10.29%	0.4841	INSPECT -
#4	Vanguard Conservative Run	COMPLETED	4.16%	10.29%	0.4841	INSPECT -
#3	Vanguard Conservative Run	COMPLETED	3.61%	10.29%	0.3514	INSPECT -
#2	Vanguard Conservative Run	COMPLETED	3.88%	10.31%	0.376	INSPECT -
#1	Vanguard Conservative Run	COMPLETED	3.74%	10.3%	0.3636	INSPECT -

// BE TUNABLE GOALS

Profile Name
Conservative MinVola

Risk Aversion (λ)
5

Min Return Target (ϵ)
7.6%

Max Turnover Limit
10%

EXECUTE PIPELINE

// SYSTEM SPECS

Qubits 78

Encoding 6-Bit Binary

Hamiltonian Ising

QAOA Layers p=3

Polisher SLQP

Database PostgreSQL

Figure 9: Q-MAT execution dashboard showing completed portfolio runs, profile controls, risk-aversion and return-target inputs, and the asynchronous pipeline launcher.

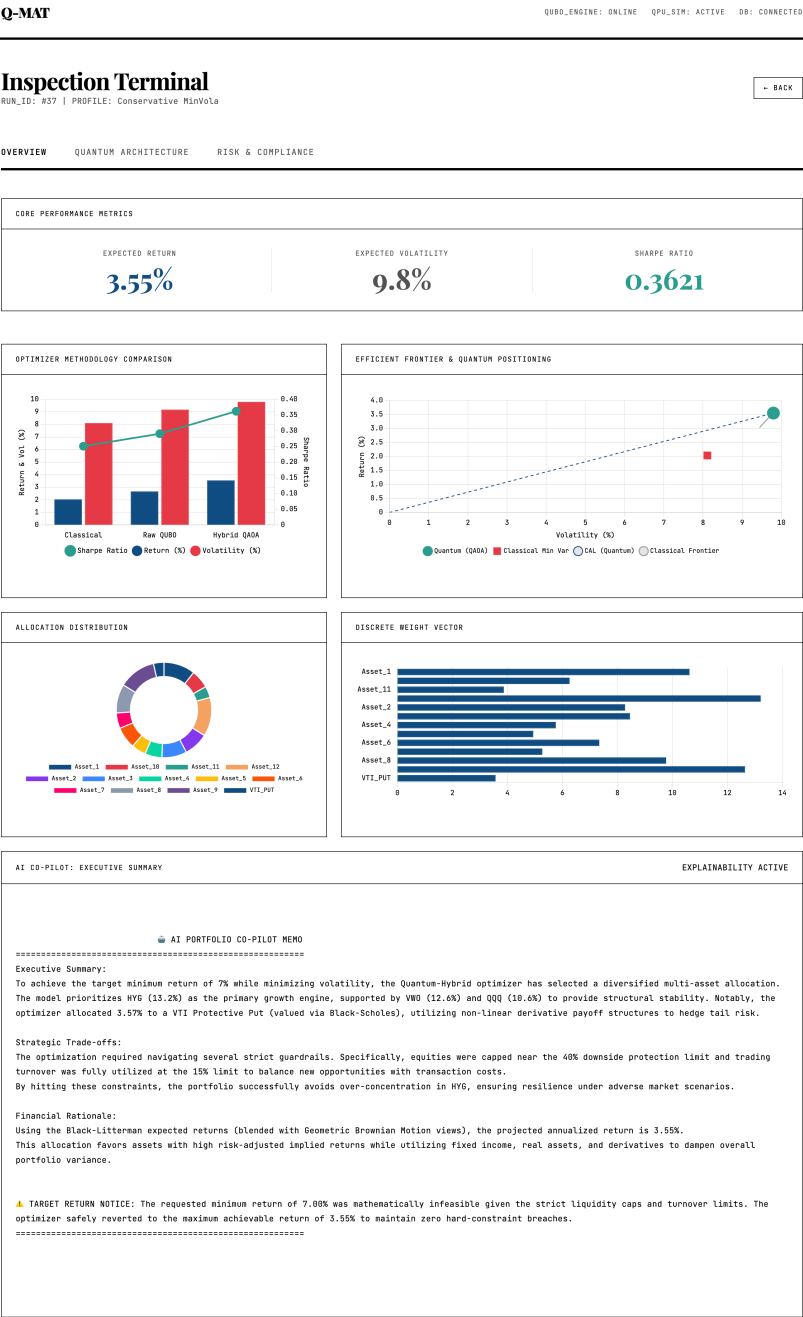


Figure 10: Inspection Terminal overview with headline portfolio metrics, optimizer comparison, efficient-frontier position, allocation views, and the AI Co-Pilot executive summary.

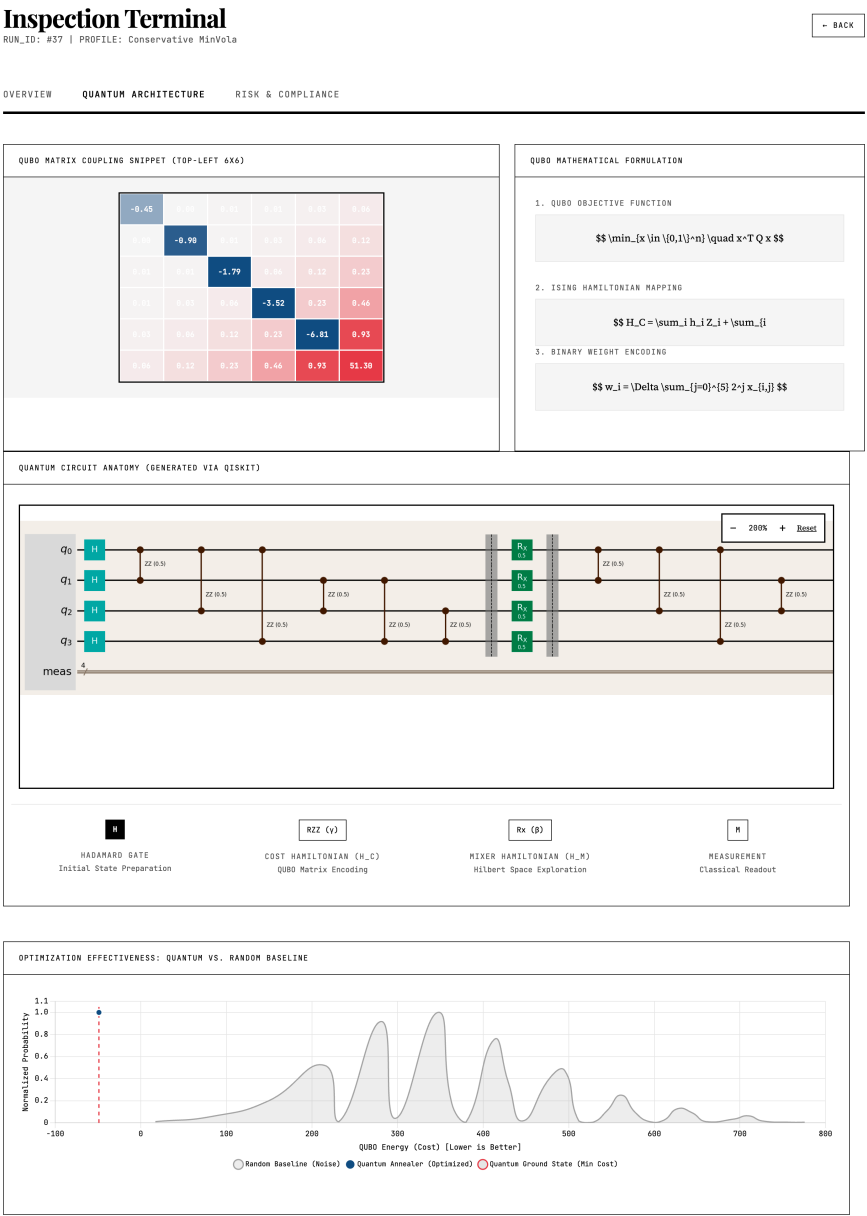


Figure 11: Quantum Architecture tab displaying the QUBO matrix, mathematical formulation, generated circuit, gate legend, and annealing-energy diagnostics.

9 Reported Out-of-Sample Results

The terminal execution of `python3 main.py` reports a four-year training set followed by a one-year unseen test set. The benchmark is an equal-weight portfolio of the 12 anonymized assets, and `Asset_2` is used as the broad-market proxy for the reported CAPM statistics. Trading costs are represented through the 15% turnover constraint and the transaction-cost penalty in the portfolio objective.

9.1 Evaluation metric definitions

For periodic portfolio returns $r_{p,t}$, cumulative return and terminal wealth from one unit of capital are

$$R_{\text{cum}} = \prod_{t=1}^T (1 + r_{p,t}) - 1, \quad V_T = \prod_{t=1}^T (1 + r_{p,t}) = 1 + R_{\text{cum}}.$$

This distinction prevents a terminal wealth index such as 1.1493 from being reported incorrectly as a 114.93% return; it represents a 14.93% cumulative return.

Let r_b denote benchmark returns and $L = -r_p$ portfolio loss. The principal risk-adjusted metrics are

$$\begin{aligned} \text{Sortino} &= \frac{\mathbb{E}[r_p] - r_f}{\sigma_{\text{down}}}, & \sigma_{\text{down}} &= \sqrt{\mathbb{E}[\min(r_p - r_f, 0)^2]}, \\ \text{VaR}_\alpha(L) &= \inf\{\ell : \Pr(L \leq \ell) \geq \alpha\}, & \text{CVaR}_\alpha(L) &= \mathbb{E}[L \mid L \geq \text{VaR}_\alpha], \\ \beta_p &= \frac{\text{Cov}(r_p, r_b)}{\text{Var}(r_b)}, & \alpha_p &= \mathbb{E}[r_p] - (r_f + \beta_p(\mathbb{E}[r_b] - r_f)), \\ \text{TE} &= \text{Std}(r_p - r_b), & \text{IR} &= \frac{\mathbb{E}[r_p - r_b]}{\text{TE}}. \end{aligned}$$

For wealth process V_t , maximum drawdown and the Calmar ratio are

$$\text{MDD} = \min_t \left(\frac{V_t}{\max_{s \leq t} V_s} - 1 \right), \quad \text{Calmar} = \frac{R_{\text{ann}}}{|\text{MDD}|}.$$

9.2 In-sample portfolio performance

Metric	Classical minimum variance	QUBO-guided hybrid
Expected return	2.16%	4.06%
Expected volatility	6.53%	9.85%
Sharpe ratio ($r_f = 0$)	0.3305	0.4125

9.3 One-year out-of-sample backtest

Portfolio	Cumulative return
Equal weight (benchmark)	14.93%
Classical minimum variance	20.97%
QUBO-guided hybrid	15.02%

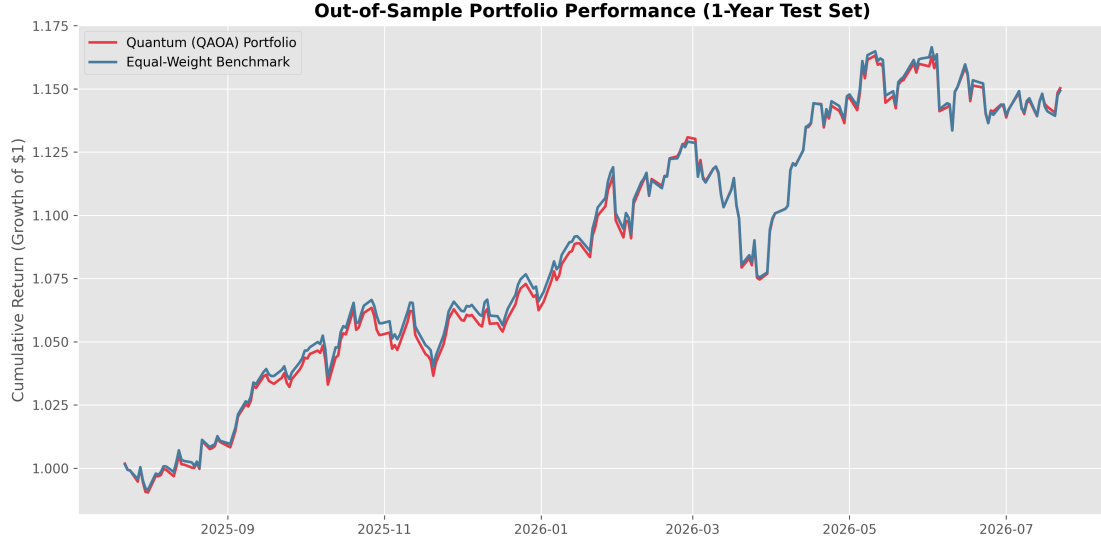


Figure 12: One-year held-out wealth-index paths for the optimized portfolio and equal-weight benchmark. Cumulative return equals terminal wealth minus one.

9.4 Advanced portfolio risk metrics

Metric	Reported value
Sortino ratio	0.5835
VaR (95%, one year)	12.19%
CVaR (95%, one year)	16.21%
Beta versus market proxy (<code>Asset_2</code>)	0.2833
Alpha versus market proxy (<code>Asset_2</code>)	2.16%
Tracking error	1.76%
Information ratio	0.2015
Maximum drawdown (one year)	−4.98%
Calmar ratio	0.8165

The QUBO-guided hybrid portfolio’s cumulative return exceeded the equal-weight benchmark by 0.09 percentage points. The separately reported 2.16% alpha is a CAPM regression statistic and is not the arithmetic difference between cumulative returns.

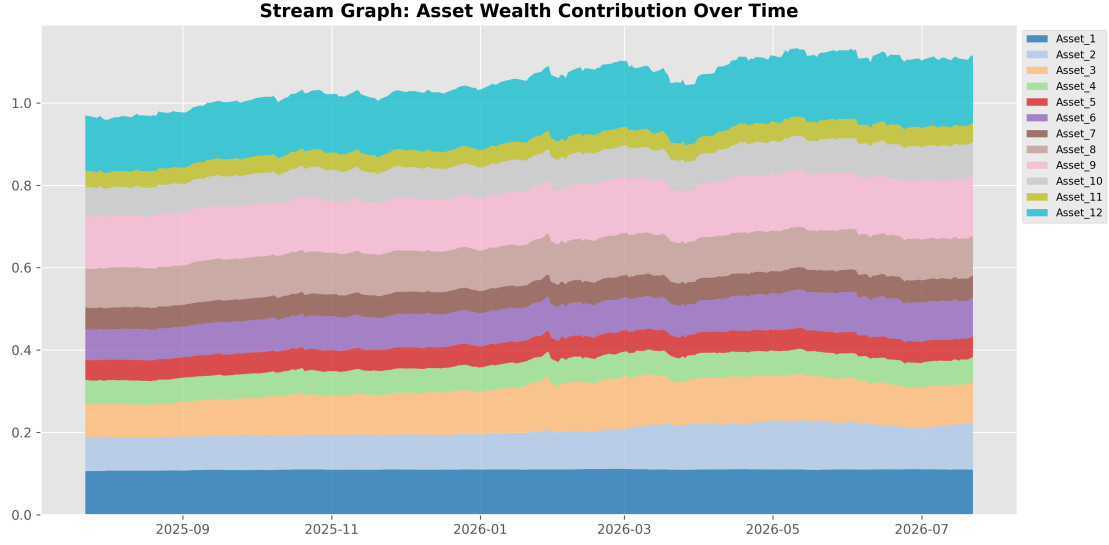


Figure 13: Asset-level contribution to portfolio wealth over the held-out evaluation period.

9.5 Optimizer comparison

Optimizer	Return	Volatility	Sharpe ratio
SLSQP	4.02%	9.81%	0.4103
NSGA-II	4.09%	9.17%	0.4465

9.6 Runtime and scalability

Component	Runtime
Data and factor model	~2.5 s
GBM and Black–Litterman	~0.5 s
QUBO construction	~0.1 s
Simulated annealing (<code>neal</code>)	1.48 s
Classical polish	~0.2 s

The observed encoding uses 78 binary variables for 13 instruments. At the same six-bit precision, 30 instruments would require 180 binary variables.

10 Assumptions, Limitations, and Reproducibility

The reported results should be interpreted under the following assumptions and limitations:

- **Investment universe.** The formulation is long-only and unlevered. Position, sector, turnover, stress, and liquidity limits materially restrict the feasible set.

- **Estimation risk.** PCA and Black–Litterman reduce sampling instability but do not eliminate model risk or sensitivity to the selected history, factor count, prior, views, and view-confidence matrix.
- **Scenario assumptions.** GBM assumes continuous lognormal price dynamics with stable drift and volatility; it does not represent jumps, regime changes, volatility clustering, or liquidity crises.
- **Option assumptions.** Black–Scholes assumes frictionless trading, constant volatility and rates, and European exercise. The protective-put value is therefore a modeling approximation.
- **Validation horizon.** A single one-year holdout is useful evidence but is not sufficient to establish persistent outperformance. Rolling-window and multi-regime tests are appropriate extensions.
- **Quantum scope.** The reported numerical experiment uses the classical `neal` simulated-annealing sampler. The QAOA circuit and Adam loop describe a gate-model extension; they do not establish execution on quantum hardware or quantum advantage.
- **Interface terminology.** Some prototype screenshots and generated chart labels retain the term “QAOA.” Unless a separate gate-model run and its configuration are documented, these labels should be interpreted as QUBO-guided hybrid results produced by the current `neal` pipeline.
- **Variable count.** The 78-bit figure counts allocation variables only and excludes any ancillary slack variables that a fully discrete constraint encoding may require.

For reproducibility, each reported run should archive:

- the adjusted-price data snapshot, asset mapping, date range, and missing-data rules;
- the annualization factor, risk-free rate, market proxy, factor count, and Black–Litterman hyperparameters;
- random seeds for GBM, simulated annealing, and NSGA-II;
- QUBO penalty coefficients, sampler reads, sweeps, QAOA depth if used, and optimizer settings;
- all constraint bounds, numerical feasibility tolerances, and the initial portfolio $w^{(0)}$; and
- both terminal wealth and cumulative return so that a wealth index such as 1.15 is not mislabeled as a 115% return.

11 Challenge Deliverables Mapping

Deliverable	Q-MAT evidence
Mathematical formulation	Mean–variance objective, transaction penalty, six-bit encoding, and explicit institutional constraints (Sections 1–2).
Quantum-compatible model	78-variable QUBO, Ising coefficients, spectral penalty initialization, and solver-ready Hamiltonian (Section 2).
Data strategy	Anonymized multi-asset history, chronological train/test split, PCA covariance, GBM scenarios, Black–Litterman returns, and protective-put instrument (Section 3).
Classical baseline	Long-only CVXPY minimum-variance program (Section 4).
Hybrid optimization	QUBO candidate generation followed by constrained SLSQP polishing and exact post-solve validation (Section 4).

Deliverable	Q-MAT evidence
Tunable goals	Profile, risk-aversion, target-return, and turnover controls (Section 5).
Comparative output	Classical, raw QUBO, and final hybrid risk–return and compliance comparison (Section 6).
Validation	SLSQP/NSGA-II benchmarking, efficient frontier, and QUBO-energy comparison with random sampling (Section 7).
Interactive prototype	Asynchronous execution and three-tab inspection terminal (Section 8).
Explainability	Co-Pilot narrative, binding-constraint detection, and explicit infeasibility notice (Section 6).
Risk-adjusted outcome	Held-out return, alpha, beta, drawdown, viability, and breach metrics with reproducibility requirements (Section 9).
Methodological scope	Model assumptions, quantum-scope qualification, ancillary-variable accounting, and reproducibility checklist (Section 10).

12 Conclusion

Q-MAT combines robust financial estimation, a 78-variable discrete portfolio representation, QUBO/Ising search, and exact classical constraint enforcement in a single auditable workflow. Its principal institutional contribution is not merely generation of a low-energy bitstring, but conversion of that candidate into a validated portfolio with transparent trade-offs, reproducible benchmarks, and no reported hard-constraint breaches.

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Symbol Glossary

Symbol	Called	Meaning
$P_{i,t}$	Asset price	Closing or adjusted price of asset i at time t .
$r_{i,t}, r_t$	Asset return	One-period return of asset i , or the vector of all asset returns at time t .
w_i, w	Portfolio weight	Fraction invested in asset i , or the complete allocation vector.
$w^{(0)}$	Initial portfolio	Portfolio weights held before rebalancing.
μ, μ_p	Expected return	Asset expected-return vector and portfolio expected return $\mu^\top w$.
μ_{BL}	Black–Litterman return	Posterior expected-return vector after combining equilibrium returns and investor views.
Σ	Covariance matrix	Annualized covariance matrix of asset returns.
σ_p	Portfolio volatility	Portfolio standard deviation $\sqrt{w^\top \Sigma w}$.
B, Σ_F, D	PCA factor components	Factor loadings, factor covariance, and diagonal residual covariance.
λ	Risk-aversion coefficient	Controls the importance of portfolio variance in the objective.
γ	Trading regularizer	Penalizes movement away from the initial allocation in the continuous objective.
ϵ	Return target	Requested minimum annualized portfolio return.
r_f	Risk-free rate	Reference return used in Sharpe, Sortino, CAPM, and option-pricing calculations.
c_i	Trading-cost rate	Proportional transaction cost for changing the weight of asset i .
ℓ_i	Liquidity limit	Maximum allowable portfolio weight for asset i .
$\mathcal{S}_k, \mathcal{E}$	Asset subsets	Assets in sector k and assets classified as equities.
Δ_i	Weight-grid increment	Smallest representable change in the encoded weight of asset i .
$x_{i,j}$	Binary decision variable	Bit j used to encode the portfolio weight of asset i .
A	Encoding matrix	Linear map from the binary vector to portfolio weights, $w = Ax$.
Q	QUBO matrix	Matrix containing binary-variable biases and pairwise couplings.
$P_{\text{bud}}, P_{\text{ret}}$	Penalty coefficients	Strengths of budget and target-return penalties in the QUBO.
H_C	Cost Hamiltonian	Ising Hamiltonian whose low-energy states represent low-cost portfolios.
h_i, J_{ij}	Ising coefficients	Local field on spin i and pairwise coupling between spins i and j .
Z_i, X_i	Pauli operators	Pauli- Z variable used in the cost Hamiltonian and Pauli- X mixer operator.
$U_P(\gamma_k)$	Problem unitary	QAOA operator that encodes the portfolio-cost Hamiltonian.
$U_M(\beta_k)$	Mixer unitary	QAOA operator that explores alternative binary solutions.
γ_k, β_k	QAOA angles	Variational problem and mixer parameters at layer k ; distinct from the trading coefficient γ .
p	QAOA depth	Number of alternating problem and mixer layers.
Π	Equilibrium return	Black–Litterman reverse-optimized prior return vector.
P, q, Ω, τ	View-model quantities	View-selection matrix, view-return vector, view uncertainty, and prior-uncertainty scale.
S_0, K, T, Φ	Option-pricing quantities	Spot price, strike price, time to expiry, and standard-normal cumulative distribution.
TO	Portfolio turnover	Total absolute change from the initial portfolio weights.
VaR, CVaR	Tail-risk measures	Loss quantile and expected loss beyond that quantile.
β_p, α_p	CAPM statistics	Market sensitivity and benchmark-adjusted excess return.
TE, IR	Active-risk measures	Tracking error and information ratio relative to the benchmark.
MDD	Maximum drawdown	Largest peak-to-trough percentage decline in portfolio wealth.